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Inequalities

An Original Olympiad Practice Set

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Original content. These 8 problems were written fresh for this set — they are **not** copied from Titu Andreescu's books or any competition. Eight original olympiad inequalities that each reward a single clean idea — AM-GM, Cauchy-Schwarz (Engel form), rearrangement, SOS, the tangent-line trick, and smoothing — rather than brute algebra. Difficulty climbs from warm-up extrema to a smoothing problem whose maximum sits on the boundary, not at the symmetric point. Every numeric answer is independently CAS-verified and every equality case is stated. Every closed-form answer was checked symbolically/numerically. Free to share for study.

Problems

1. DIFFICULTY 1/5 · AM-GM

For $x > 0$, determine the minimum value of

$$f(x) = 9x + \frac{4}{x},$$

and state where it is attained.

2. DIFFICULTY 2/5 · AM-GM, CAUCHY-SCHWARZ

Let $a, b, c > 0$. Prove that

$$(a+b+c)\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right) \;\geq\; 9,$$

and determine all equality cases.

3. DIFFICULTY 2/5 · REARRANGEMENT, SOS

Let $a, b, c > 0$. Prove that

$$a^3+b^3+c^3 \;\geq\; a^2b+b^2c+c^2a,$$

and find all cases of equality.

4. DIFFICULTY 3/5 · CAUCHY-SCHWARZ, AM-GM

Let $a, b, c > 0$. Prove that

$$(a+b+c)\left(\frac{1}{a+b}+\frac{1}{b+c}+\frac{1}{c+a}\right) \;\geq\; \frac{9}{2},$$

and describe the equality case.

5. DIFFICULTY 3/5 · CAUCHY-SCHWARZ, ENGEL FORM

Let $a, b, c > 0$ with $a+b+c=1$. Prove that

$$\frac{a^2}{a+b}+\frac{b^2}{b+c}+\frac{c^2}{c+a} \;\geq\; \frac{1}{2},$$

and state when equality holds.

6. DIFFICULTY 4/5 · TANGENT LINE, SOS

Let $a, b, c > 0$ with $a+b+c=3$. Prove that

$$\frac{a}{a^2+2}+\frac{b}{b^2+2}+\frac{c}{c^2+2} \;\leq\; 1,$$

and determine the equality case.

7. DIFFICULTY 4/5 · CAUCHY-SCHWARZ, ENGEL FORM, AM-GM

Let $a, b, c > 0$ with $a + b + c = 3$. Prove that

$$\frac{a^2}{a + 2bc} + \frac{b^2}{b + 2ca} + \frac{c^2}{c + 2ab} \ \geq \ 1,$$

and find the equality case.

8. DIFFICULTY 5/5 · SMOOTHING, AM-GM, BOUNDARY ANALYSIS

Let $a, b, c \geq 0$ with $a + b + c = 3$. Find the maximum value of

$$E = ab + bc + ca - abc,$$

and determine all triples (a, b, c) achieving it.

Solutions

1. Answer: 12, attained at $x = \frac{2}{3}$

Both terms are positive, so by AM-GM applied to the two quantities $9x$ and $\frac{4}{x}$,

$$9x + \frac{4}{x} \;\geq\; 2\sqrt{(9x)\cdot\frac{4}{x}} = 2\sqrt{36} = 12.$$

The product $(9x)\cdot\frac{4}{x} = 36$ is constant in x, which is exactly what makes AM-GM yield a fixed bound here. Equality in AM-GM holds iff the two terms are equal: $9x = \frac{4}{x}$, i.e. $x^2 = \frac{4}{9}$, so $x = \frac{2}{3}$ (taking the positive root). Then $f(\frac{2}{3}) = 6 + 6 = 12$. Hence the minimum is $\boxed{12}$, achieved at $x = \frac{2}{3}$.

Second method. Calculus: $f'(x) = 9 - \frac{4}{x^2} = 0 \Rightarrow x^2 = \frac{4}{9} \Rightarrow x = \frac{2}{3}$. Since $f''(x) = \frac{8}{x^3} > 0$ for $x > 0$, this is the global minimum, with $f(\frac{2}{3}) = 12$.

2. Answer: Proof (minimum value 9, equality iff $a = b = c$)

Apply AM-GM to each factor separately. For the three positive numbers a, b, c,

$$a + b + c \;\geq\; 3\sqrt[3]{abc},$$

and for their reciprocals,

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \;\geq\; 3\sqrt[3]{\frac{1}{abc}}.$$

Multiplying the two inequalities (all quantities positive) gives

$$(a+b+c)\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) \;\geq\; 9\sqrt[3]{abc}\cdot\sqrt[3]{\frac{1}{abc}} = 9.$$

Equality requires equality in both AM-GM steps, i.e. $a = b = c$. Conversely $a = b = c$ gives $(3a)(3/a) = 9$. Hence the minimum is 9, attained exactly when $a = b = c$.

Second method. Cauchy-Schwarz: $(\sum a)\left(\sum \frac{1}{a}\right) \geq \left(\sum \sqrt{a}\cdot\frac{1}{\sqrt{a}}\right)^2 = (1+1+1)^2 = 9$, with equality iff the vectors $(\sqrt{a}, \sqrt{b}, \sqrt{c})$ and $(1/\sqrt{a}, 1/\sqrt{b}, 1/\sqrt{c})$ are proportional, i.e. $a = b = c$.

3. Answer: Proof (equality iff $a = b = c$)

By the rearrangement inequality, the sum $\sum a^2 \cdot a$ pairing the equally-sorted sequences (a^2, b^2, c^2) and (a, b, c) is the largest among all permutation-pairings, since for positive reals $a \le b \le c \iff a^2 \le b^2 \le c^2$, so the two sequences are similarly ordered. The cyclic pairing $a^2b + b^2c + c^2a$ is one such (generally non-identical) permutation, hence

$$a^3 + b^3 + c^3 = a^2{\cdot}a + b^2{\cdot}b + c^2{\cdot}c \;\ge\; a^2b + b^2c + c^2a.$$

Equality in rearrangement (with strictly ordered sequences) forces the permutation to be identity, which happens precisely when the sequences are constant, i.e. $a = b = c$.

Second method. SOS via weighted AM-GM term-by-term: $\frac{2a^3+b^3}{3} \ge a^2b$ by AM-GM on $\{a^3, a^3, b^3\}$ (indeed $\frac{2a^3+b^3}{3} - a^2b = \frac{(a-b)^2(2a+b)}{3} \ge 0$). Summing the three cyclic copies,

$$\frac{2a^3+b^3}{3} + \frac{2b^3+c^3}{3} + \frac{2c^3+a^3}{3} = a^3 + b^3 + c^3 \;\ge\; a^2b + b^2c + c^2a,$$

with equality iff $a = b$, $b = c$, $c = a$, i.e. $a = b = c$.

4. Answer: Proof (minimum value $\frac{9}{2}$, equality iff $a = b = c$)

Let $x = b + c,\; y = c + a,\; z = a + b$. These are positive, and $x + y + z = 2(a + b + c)$, so $a + b + c = \frac{x+y+z}{2}$. The claim becomes

$$\frac{x+y+z}{2}\left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z}\right) \ge \frac{9}{2}, \quad \text{i.e.} \quad (x+y+z)\left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z}\right) \ge 9.$$

This last inequality is exactly INE-2 applied to x, y, z (or directly Cauchy-Schwarz: $(x+y+z)(\frac{1}{x} + \frac{1}{y} + \frac{1}{z}) \ge (1+1+1)^2 = 9$). Dividing by 2 gives the result. Equality requires $x = y = z$, i.e. $b + c = c + a = a + b$, which gives $a = b = c$; then the left side equals $(3a) \cdot \frac{3}{2a} = \frac{9}{2}$.

Second method. Direct Engel form (Cauchy-Schwarz): $\sum \frac{1}{a+b} = \sum \frac{1^2}{a+b} \ge \frac{(1+1+1)^2}{(a+b)+(b+c)+(c+a)} = \frac{9}{2(a+b+c)}$. Multiplying by $a + b + c$ gives $\ge \frac{9}{2}$, equality iff all denominators equal, i.e. $a = b = c$.

5. Answer: Proof (minimum value $\frac{1}{2}$, equality iff $a = b = c = \frac{1}{3}$)

Apply the Cauchy-Schwarz inequality in Engel (Titu) form to the three fractions:

$$\frac{a^2}{a+b} + \frac{b^2}{b+c} + \frac{c^2}{c+a} \ge \frac{(a+b+c)^2}{(a+b)+(b+c)+(c+a)}.$$

The denominator equals $2(a + b + c) = 2$, and the numerator is $(a + b + c)^2 = 1$. Therefore the sum is $\ge \frac{1}{2}$. Equality in the Engel form holds iff

$$\frac{a}{a+b} = \frac{b}{b+c} = \frac{c}{c+a},$$

which (combined with $a + b + c = 1$) forces $a = b = c = \frac{1}{3}$; indeed each fraction then equals $\frac{1/9}{2/3} = \frac{1}{6}$, summing to $\frac{1}{2}$.

Second method. "Subtract the trivial part." Write $\frac{a^2}{a+b} = a - \frac{ab}{a+b}$. Summing, the sum equals $(a + b + c) - \sum \frac{ab}{a+b} = 1 - \sum \frac{ab}{a+b}$. Since $\frac{ab}{a+b} \le \frac{a+b}{4}$ (AM-GM: $4ab \le (a + b)^2$), we get $\sum \frac{ab}{a+b} \le \frac{1}{4} \sum(a + b) = \frac{2(a+b+c)}{4} = \frac{1}{2}$, so the original sum $\ge 1 - \frac{1}{2} = \frac{1}{2}$. Equality iff $a = b$, $b = c$, $c = a$, i.e. $a = b = c = \frac{1}{3}$.

6. Answer: Proof (maximum value 1, equality iff $a = b = c = 1$)

Use the tangent-line trick. The symmetric extremum is expected at $a = b = c = 1$, where each term equals $\frac{1}{3}$. Consider $g(t) = \dfrac{t}{t^2+2}$ and its tangent line at $t = 1$. Since $g(1) = \frac{1}{3}$ and $g'(t) = \dfrac{2-t^2}{(t^2+2)^2}$ gives $g'(1) = \frac{1}{9}$, the tangent line is $\ell(t) = \frac{1}{3} + \frac{1}{9}(t-1) = \dfrac{t+2}{9}$. We claim

$$\frac{t}{t^2+2} \le \frac{t+2}{9} \qquad \text{for all } t > 0.$$

Cross-multiplying (positive denominators): $(t+2)(t^2+2) - 9t = t^3 + 2t^2 - 7t + 4$. Factoring, $t^3 + 2t^2 - 7t + 4 = (t-1)^2(t+4)$, which is ≥ 0 for all $t > 0$ since $(t-1)^2 \ge 0$ and $t + 4 > 0$. This proves the pointwise bound. Summing over a, b, c and using $a + b + c = 3$,

$$\sum \frac{a}{a^2+2} \le \sum \frac{a+2}{9} = \frac{(a+b+c)+6}{9} = \frac{3+6}{9} = 1.$$

Equality requires $(a-1)^2 = (b-1)^2 = (c-1)^2 = 0$, i.e. $a = b = c = 1$, which indeed gives sum $= 1$.

Second method. Pure SOS form of the same bound, with no calculus and no concavity assumptions: for each $a > 0$,

$$\frac{a+2}{9} - \frac{a}{a^2+2} = \frac{(a-1)^2(a+4)}{9\,(a^2+2)} \ge 0.$$

Hence $\dfrac{a}{a^2+2} \le \dfrac{a+2}{9}$ term by term, and summing with $a+b+c = 3$ gives $\sum \frac{a}{a^2+2} \le \frac{(a+b+c)+6}{9} = 1$, with equality iff each squared factor vanishes, i.e. $a = b = c = 1$. (Note: a naive Jensen/concavity argument fails here because g is only concave on $(0, \sqrt{6}\,]$ while a variable can lie in $(\sqrt{6}, 3)$; the tangent-line / SOS bound above is the correct elementary route.)

7. Answer: Proof (minimum value 1, equality iff $a = b = c = 1$)

By the Cauchy-Schwarz (Engel/Titu) form,

$$\sum \frac{a^2}{a+2bc} \ge \frac{(a+b+c)^2}{(a+2bc)+(b+2ca)+(c+2ab)} = \frac{(a+b+c)^2}{(a+b+c)+2(ab+bc+ca)}.$$

Let $p = a+b+c = 3$ and $q = ab+bc+ca$. The bound becomes $\dfrac{9}{3+2q}$. It remains to show $\dfrac{9}{3+2q} \ge 1$, i.e. $3 + 2q \le 9$, i.e. $q \le 3$. But for any reals, $(a+b+c)^2 \ge 3(ab+bc+ca)$ (equivalently $\frac{1}{2}\sum(a-b)^2 \ge 0$), so $q \le \frac{p^2}{3} = \frac{9}{3} = 3$. Hence the sum is $\ge \frac{9}{3+2\cdot 3} = \frac{9}{9} = 1$. Equality requires equality in Cauchy-Schwarz, namely $\frac{a}{a+2bc} = \frac{b}{b+2ca} = \frac{c}{c+2ab}$, together with $q = 3$; both force $a = b = c = 1$, which gives each term $\frac{1}{3}$ summing to 1.

Second method. The decisive insight is that the messy denominators collapse under Cauchy-Schwarz to the single symmetric quantity $p + 2q$: $(a+2bc)+(b+2ca)+(c+2ab) = (a+b+c)+2(ab+bc+ca) = p+2q$. After the Engel step the whole problem reduces to the elementary $q \le \frac{p^2}{3} = 3$, i.e. $ab+bc+ca \le \frac{1}{3}(a+b+c)^2$, which is just $\frac{1}{2}\sum(a-b)^2 \ge 0$ with equality iff $a = b = c$. This is the cleanest closing bound; no per-term denominator manipulation is needed.

8. Answer: $\dfrac{9}{4}$, attained exactly at the permutations of $\left(\frac{3}{2}, \frac{3}{2}, 0\right)$

The key insight is that the maximum is NOT at the symmetric point: at $a = b = c = 1$, $E = 3 - 1 = 2 < \frac{9}{4}$. It lives on the boundary. We use smoothing. Among a, b, c let $c = \min$; since the average is 1, $c \le 1$. Fix c and the partial sum $s = a + b = 3 - c$, and view E as a function of a, b:

$$E = ab + c(a + b) - abc = ab(1 - c) + cs.$$

Because $c \le 1$, the coefficient $1 - c \ge 0$, so E is maximized by making ab as large as possible. With $a + b = s$ fixed, $ab \le \left(\frac{s}{2}\right)^2$ by AM-GM, attained at $a = b = \frac{s}{2}$. Thus for the optimum we may take $a = b = \frac{3-c}{2}$, reducing E to a single variable:

$$E(c) = \left(\tfrac{3-c}{2}\right)^2 (1 - c) + c(3 - c), \qquad c \in [0, 1].$$

Expanding,

$$E(c) = \tfrac{1}{4}(3 - c)^2(1 - c) + c(3 - c) = \tfrac{1}{4}\left(-c^3 + 3c^2 - 3c + 9\right).$$

Its derivative is

$$E'(c) = \tfrac{1}{4}\left(-3c^2 + 6c - 3\right) = -\tfrac{3}{4}\,(c - 1)^2 \;\le\; 0,$$

a negative multiple of a perfect square. Hence $E(c)$ is (weakly) decreasing on $[0, 1]$, and strictly decreasing except at the isolated point $c = 1$; its maximum on $[0, 1]$ is therefore at $c = 0$:

$$E(0) = \tfrac{1}{4}(0 - 0 - 0 + 9) = \frac{9}{4}.$$

At $c = 0$, $a = b = \frac{3}{2}$. Therefore $\max E = \dfrac{9}{4}$, attained exactly at the permutations of $\left(\frac{3}{2}, \frac{3}{2}, 0\right)$ (the smoothing chain shows no interior point can match it).

Second method. Reduce to two variables via the constraint, then a sharp AM-GM. Setting $c = 0$ is suggested by the boundary insight; on the face $c = 0$ we maximize ab with $a + b = 3$, giving $ab \le \frac{9}{4}$ at $a = b = \frac{3}{2}$, so $E = ab \le \frac{9}{4}$ there. That no interior triple beats this is exactly the content of the smoothing computation: writing $E = q - r$ with $q = ab + bc + ca,\ r = abc$, the symmetric bound $E \le q \le \frac{p^2}{3} = 3$ is too weak — which is why the maximum is forced onto the boundary, where some variable (hence $r = 0$) vanishes. Confirming the maximum $\frac{9}{4}$ at $\left(\frac{3}{2}, \frac{3}{2}, 0\right)$ and permutations.